MIL-A-18065(SHIPS)
22 July 1954

MILITARY SPECIFICATION

ADHESIVES, HIGH INITIAL BOND

1. SCOPE

1.1 Scope. - This specification covers high-initial bond adhesive for use in securing corkboard insulation to metal surfaces in submarines.

1.2 Classification. - Adhesive shall be of the following classes, as specified (see 6.2):

Class 1 - Fast-setting.
Class 2 - Slow-setting.

2. APPLICABLE DOCUMENTS

2.1 The following specifications and standards, of the issue in effect on date of invitation for bids, form a part of this specification:

SPECIFICATIONS

FEDERAL
HH-C-561 - Cork; Compressed (Corkboard) (for Thermal Insulation).
VV-L-791 - Lubricants, Liquid Fuels, and Related Products; Methods of Inspection, Sampling and Testing.

MILITARY
MIL-C-124 - Containers (Cans, Pails, and Drums), Metal (for Other Than Substitute Items).

NAVY DEPARTMENT
General Specifications for Inspection of Materials.

STANDARDS -

MILITARY
MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.
MS15153 - Containers, (Cans, Pails, and Drums), Steel, Filled; Inspection Standard.

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring agency or as directed by the contracting officer.)

2.2 Other publications. - The following document forms a part of this specification: Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply.

CONSOLIDATED CLASSIFICATION COMMITTEE
Consolidated Freight Classification Rules.

(Application for copies should be addressed to the Consolidated Classification Committee, 202 Chicago Union Station, Chicago 6, Ill.)

FED. SUP. CLASS
8040

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3. REQUIREMENTS

3.1 Qualification.- Adhesive furnished under this specification shall be a product which has been tested and has passed the qualification tests specified herein (see 6.3).

3.2 Material.- Adhesive, in the condition in which delivered, shall be suitable and effective for the purpose intended without heating or additions of other ingredients, and shall be free from grit, lumps, and skins, and shall not gel, liver, settle hard or otherwise deteriorate when stored for a period of one year in airtight containers. The adhesive shall be free from all ingredients which may effect the serviceability or have a deleterious effect on the metal or insulation to which the adhesive is applied. The adhesive shall be free from chlorinated solvents, benzol, or any other toxic ingredients which would adversely affect the health of workmen using them.

3.3 Protection against corrosion.- Adhesive shall protect bare steel surfaces from corrosion when tested as specified in 4.3.1.

3.4 Flexibility.- Adhesive shall not chip, scale, crack or otherwise leave the surface of the tin strip when tested as specified in 4.3.2.

3.5 Consistency.- The cone penetration of adhesive shall be not less than 25 nor more than 35 millimeters (mm.) (see 4.3.3).

3.6 Shear strength.- The shear strength of adhesive shall be sufficient to support a one-pound weight for a period of 72 hours (see 4.3.4).

3.7 Initial bond adhesive strength.- The initial bond adhesive strength of the adhesive shall be not less than 60 pounds per square foot (see 4.3.5).

3.8 Class 1.-

3.8.1 General.- Adhesive shall be a fast-setting, high-initial bond, volatile-solvent-type adhesive suitable for securing compressed cork conforming to Specification HH-C-561 to bare steel surfaces in easily ventilated compartments (see 6.1).

3.8.2 Ultimate adhesive strength.-

3.8.2.1 The average ultimate adhesive strength shall be not less than the values shown in table I.

Table I - Ultimate adhesive strength.

	Method of test, paragraph			
	4.3.6.3.1	4.3.6.3.2	4.3.6.3.3	4.3.6.3.4
Ultimate adhesive strength, pounds per square foot, minimum	1,500	1,000	1,500	1,000

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3.8.2.2 Adhesive shall be considered as conforming to the strength requirements specified in 3.8.2.1 when failure occurs in the corkboard at a strength lower than than specified in table I for the adhesive.

3.9 Class 2.-

3.9.1 General.- Adhesive shall be a nonhardening, high-initial-bond, high-flash point, nontoxic-solvent-type adhesive suitable for securing compressed cork conforming to Specification HH-C-561 to bare steel surfaces in confined, hard-to-ventilate compartments (see 6.1).

3.9.2 Ultimate adhesive strength.-

3.9.2.1 The average ultimate adhesive strength shall be not less than the values shown in table II.

Table II - Ultimate adhesive strength.

	Method of test, paragraph			
	4.3.6.3.1	4.3.6.3.2	4.3.6.3.3	4.3.6.3.4
Ultimate adhesive strength, pounds per square foot, minimum	1,500	250	750	750

3.9.2.2 Adhesive shall be considered as conforming to the strength requirements specified in 3.9.2.1 when failure occurs in the corkboard at a strength lower than that specified in table II for the adhesive.

3.9.3 Flask point.- The flash point of the adhesive shall be not less than 120° Fahrenheit (F.) (see 4.3.7).

4. QUALITY ASSURANCE PROVISIONS

4.1 Sampling.-

4.1.1 Lot.- For purposes of sampling a lot shall consist of all cement from one production batch offered for delivery at one time.

4.1.2 Sampling for inspection of filled containers (at the place of manufacture).- A random sample of filled containers shall be selected by the Government inspector in accordance with Standard MIL-STD-105 at inspection level I and Acceptable Quality Level = 2.5 percent defective for the inspection specified in 4.2.1.

4.1.3 Sampling for lot acceptance tests (at a Government laboratory).- From each lot the Government inspector shall select two representative 2-quart samples for the tests specified in 4.2.2. Unless otherwise specified in the contract or order, the samples so selected shall be forwarded by the Government inspector to the U.S. Naval Engineering Experiment Station, Annapolis, Md.

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4.2 Inspection and tests.-

4.2.1 Inspection.- Each filled container in the sample shall be examined to verify compliance with this specification. Standard MS15153 shall be used for classification of defects. Any container in the sample having one or more defects or under required fill shall be rejected, and if the number of defective containers in any sample exceeds the acceptance number for the appropriate sampling plan of Standard MIL-STD-105, the lot represented by the sample shall be rejected. Rejected lots may be resubmitted for Government inspection provided the contractor has removed or repaired all nonconforming containers.

4.2.2 Lot acceptance tests.- The sample selected in accordance with 4.1.3 shall be subjected to any of the tests specified in 4.3 deemed necessary by the Government laboratory to determine that the specimen conforms with that given qualification.

4.2.2.1 Action in case of failure.- If the sample is found not to be in conformance with this specification, the lot which it represents shall be rejected. A rejected lot may be resubmitted for Government inspection only after the manufacturer, having been informed of the reasons for rejection, has so reworked the entire lot as to remove or correct all nonconforming material.

4.3 Test procedures.-

4.3.1 Protection against corrosion.- Ground steel strips 1/8 by 1 by 6 inches shall be dipped in the adhesive and allowed to dry for 24 hours. At the end of this time the strips shall be subjected to a 4-percent salt solution spray for 72 hours at 100°F. The strips shall be removed from the spray and examined for penetration of the salt spray and corrosion of the metal surfaces.

4.3.2 Flexibility.-

4.3.2.1 Apparatus.- The apparatus shall consist of a 8 by 12 by 0.011 inch panel of bright tinplate, a film applicator with a clearance of 0.02 inch and a gap of sufficient size to make a film 3 to 6 inches wide, and a steel mandrel 1/4 inch in diameter and 6 inches long supported at each end.

4.3.2.2 Procedure.- The panel shall be placed on a flat surface and a film of the adhesive 12 inches long and 3 to 6 inches wide shall be drawn down using the film applicator. The film shall be allowed to dry at room temperature (77° ± 5°F.) for 24 hours with the panel in horizontal position. Four 1-1/2 by 4 inch strips shall be cut from the coated panel. Two of the strips shall be placed in an air-circulating oven at 212° to 220°F. for 75 minutes. The other two strips shall be immersed in fresh water at a temperature of 75° ± 5°F. for 48 hours. The strips shall be removed from the oven and the water, allowed to dry and cool to room temperature, and then bent rapidly through a 180° arc over the surface of the mandrel. Then the strips shall be inspected for chips, scales and cracks at the bend. Slight surface cracking shall be disregarded, but deep cracks which expose bare metal shall be considered as failure to pass the test.

4.3.3 Consistency.-

4.3.3.1 Apparatus.- The apparatus used shall be as specified in method 311 of Specification VV-L-791. The total weight of the cone and rod shall be 150 grams.

4.3.3.2 Procedure.- A thin film of water shall be floated on top of the test adhesive during the test to prevent nonconformity due to evaporation or skinning. The open cup portion of the grease worker (see method 311 of Specification VV-L-791) shall be completely filled with the well-mixed sample at a temperature of 77° ± 1°F. A straight edge shall be drawn across the surface of the sample to remove any excess material and to present a smooth surface. The apparatus shall be leveled and the plunger lowered until the tip of the penetrometer cone just touches the surface of the sample. The scale shall then be adjusted so that the scale actuating device is in contact with the top of the rod holding the penetrometer cone and the scale reading recorded. The plunger shall be released suddenly and kept released

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for 5 seconds. The scale actuating device shall be moved until it is again in contact with the top of the rod holding the penetrometer cone, and the scale reading recorded. The penetration is the difference between the two readings. Five tests shall be made and the average reported. The sample shall be smoothed over before each test.

4.3.4 Shear strength.- A 6 by 6 by 1-inch section of corkboard insulation conforming to Specification HH-C-561 shall be trowel-coated on one face with the adhesive and allowed to set for 5 minutes. The cork then shall be applied with a rotary sliding motion to the vertical, mill scale surface of a steel plate. A 1-pound weight shall be immediately suspended from the specimen with a light sheet metal yoke framed around the cork adjacent to the adhesive line. The specimen shall be observed closely for slippage of the cork and shear of the cement. If no initial shear occurs, the specimen shall be observed over a 72-hour period for possible failure.

4.3.5 Initial-bond adhesive strength.-

4.3.5.1 Apparatus.- The apparatus shall consist of two aluminum discs 7-1/2 by 5-7/16 inches in diameter, by 1/4 inch thick, an aluminum ring 5-9/16-inch diameter by 7-1/2-inch outer diameter by 1/4 inch thick, and two threaded studs with wingnuts to fit holes drilled in the larger disc and ring to hold them centered, parallel and 1 inch apart.

4.3.5.2 A 6 by 6 by 1-inch specimen of corkboard insulation conforming to Specification HH-C-561 shall be trowel-coated on one side with the test adhesive and laid coated side up on the larger disc. The ring then shall be bolted in place on the coated side, and the assembly shall be dried at room temperature (77° ± 5°F.) for 5 minutes. Then the smaller disc shall be placed in the hollow of the ring, its center concentric to that of the ring, using a rotary sliding motion to insure full contact between the disc and the adhesive. Hooks shall be secured in centered holes on the outer faces of the two discs. The adhesion of the smaller disc to the adhesive shall be determined immediately in a tensile machine, the lower jaw of which shall move at a uniform rate of 12 ± 1/2 inches per minute. Initial bond adhesive strength shall be the average of three determinations.

4.3.6 Ultimate adhesive strength.-

4.3.6.1 Apparatus.- Two steel discs 4 inches in diameter and 1/2 inch thick shall be drilled and tapped in the center to receive a 1/2 inch bolt. The tapped holes shall not be drilled entirely through the discs. Each bolt shall be drilled radially at the outer end to receive a swivel joint steel pin for securing to the testing machine.

4.3.6.2 Preparation.- The adhesive shall be applied with a serrated trowel to the surface of the two steel discs and allowed to set for 5 minutes. The coated steel discs shall then be applied to the two surfaces of a 4-inch compressed cork disc, 1 inch thick, in such a manner that the cork disc will be sandwiched between the steel discs.

4.3.6.3 Procedure.-

4.3.6.3.1 Three specimens shall be prepared as specified in 4.3.6.2. They shall be dried for 72 hours at room temperature (77° ± 5°F.). Then they shall be pulled to rupture in a testing machine, the lower jaw of which moves at a uniform rate of 12 ± 1/2-inch per minute.

4.3.6.3.2 Three specimens shall be prepared as specified in 4.3.6.2. They shall be dried for 72 hours at room temperature (77° ± 5°F.) and for 2 hours at 158°F. Then they shall be pulled to rupture in a testing machine, the lower jaw of which moves at a uniform rate of 12 ± 1/2 inch per minute.

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4.3.6.3.3 Three specimens shall be prepared as specified in 4.3.6.2. They shall be dried for 72 hours at room temperature and immersed in tap water for 24 hours. Then they shall be pulled to rupture in a testing machine, the lower jaw of which moves at a uniform rate of 12 ± 1/2-inch per minute.

4.3.6.3.4 Three specimens shall be prepared as specified in 4.3.6.2. The adhesive and cork discs shall be cooled to 20°F. for 24 hours prior to preparation of the specimens. The specimens shall be prepared singly and rapidly to avoid condensation of moisture on the steel discs. They shall be dried for 72 hours at 20°F. Then they shall be pulled to rupture in a testing machine, the lower jaw of which moves at a uniform rate of 12 ± 1/2 inch per minute.

4.3.7 Flash point.- The flash point shall be determined in accordance with method 1103 of Specification VV-L-791.

4.3.8 Storagability.- The adhesive shall be stored for a period of one year in airtight containers. At the end of this period, the adhesive shall be examined to determine conformance with 3.2.

4.4 Qualification tests at a Government laboratory.- Qualification tests shall be conducted at a Government laboratory designated by the Bureau of Ships. These tests shall consist of the tests specified in 4.3.

4.5 Inspection procedures.- For Naval purchases, the general inspection procedures shall be in accordance with General Specifications for Inspection of Materials.

5. PREPARATION FOR DELIVERY

5.1 Packaging.-

5.1.1 For domestic shipment - immediate use.- Packaging shall be in accordance with the manufacturer's commercial practice.

5.1.2 For domestic shipment and storage or overseas shipment.- Adhesive shall be packaged in 1-gallon multiple-friction-top cans or 5-gallon lug cover steel pails as specified (see 6.1). One-gallon containers shall conform to type IV of Specification MIL-C-124 and 5-gallon containers shall conform to type VI of Specification MIL-C-124.

5.2 Packing.-

5.2.1 For domestic shipment - immediate use.- Adhesive packaged as specified in 5.1.1 shall be packed in containers in a manner to insure acceptance and safe delivery at destination. Containers shall comply with the Consolidated Freight Classification Rules or other regulations as applicable to the mode of transportation.

5.2.2 For domestic shipment and storage or overseas shipment.- Adhesive packaged as specified in 5.1.2 shall be packed in accordance with Specification MIL-C-124 and appendix thereto.

5.3 Marking.- In addition to any special marking required by the contract or order, marking of unit and shipping containers shall conform to Standard MIL-STD-129.

6. NOTES

6.1 Class 1 (fast setting) adhesive.- Class 1 (fast-setting) adhesive is intended for use in areas where fire and personnel hazards can be eliminated by ventilation.

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6.1.1 Class 2 (slow-setting) adhesive.- Class 2 (slow-setting) adhesive is intended for use in dead air spaces where use of class 1 adhesive would present a hazard.

6.2 Ordering data.- Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Class required (see 1.2).
(c) Size of container (see 5.1.2).
(d) Whether preparation for domestic shipment - immediate use; domestic shipment and storage, or overseas shipment is required. (See 5.1 and 5.2.)

6.3 In the procurement of products requiring qualification, the right is reserved to reject bids on products that have not been subjected to the required tests and found satisfactory for inclusion on the Military Qualified Products List. The attention of suppliers is called to this requirement, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government, tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. Information pertaining to qualification of products covered by this specification may be obtained from the Chief of the Bureau of Ships, Navy Department, Washington 25, D. C.

Patent notice.- When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

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